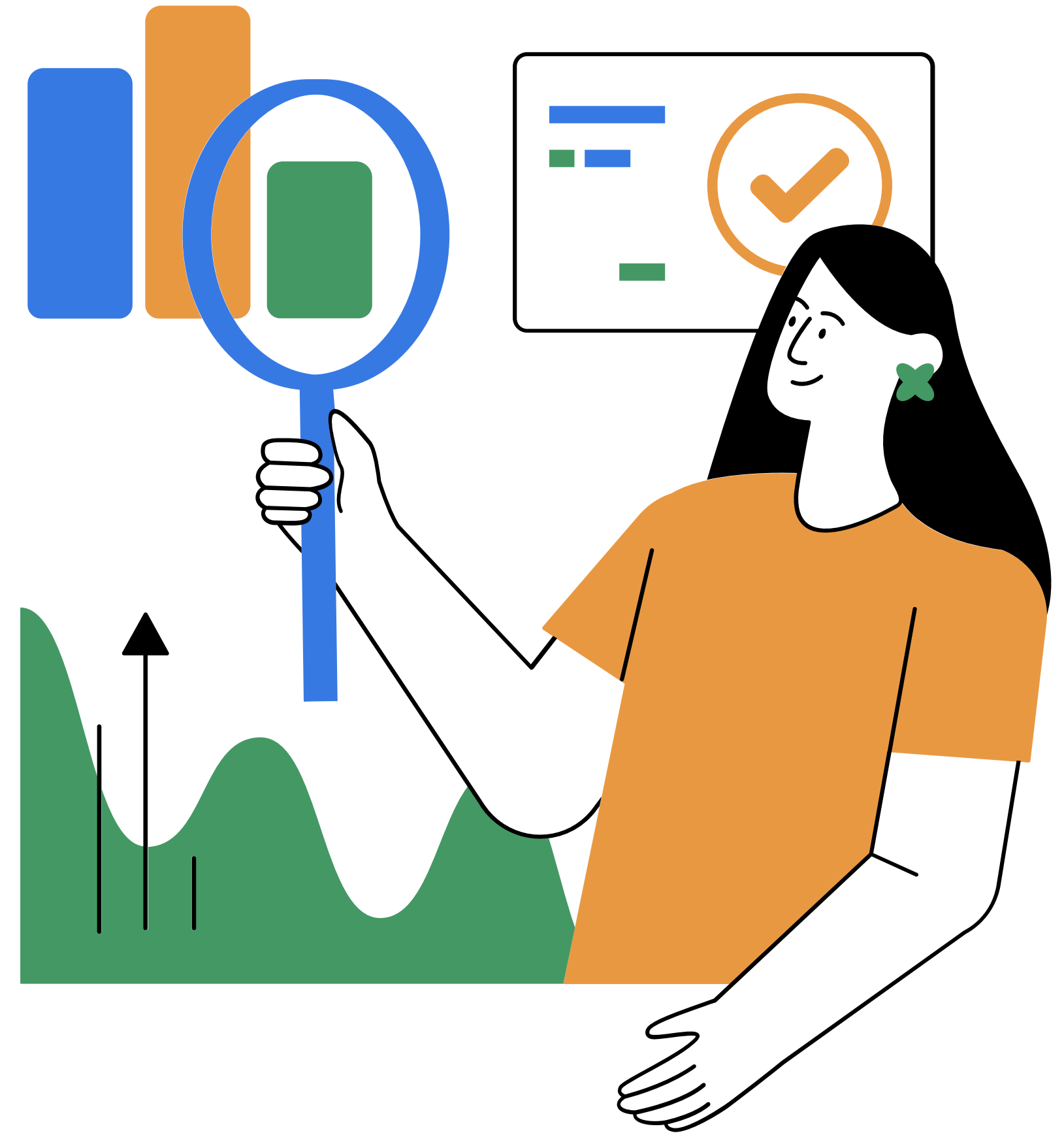


DVA CAPSTONE

Group - 6

- Omkar Hadole
- Agrim Malhotra
- Narendra Singh
- Rohan Singh
- Ritik Ranjan
- vidhi singhal



Executive Summary



Average Income
₹41,585.5

Average Expense
₹12,620.32

Average Saving
₹4,982.88

Saving rate
9.80%

Key Findings:

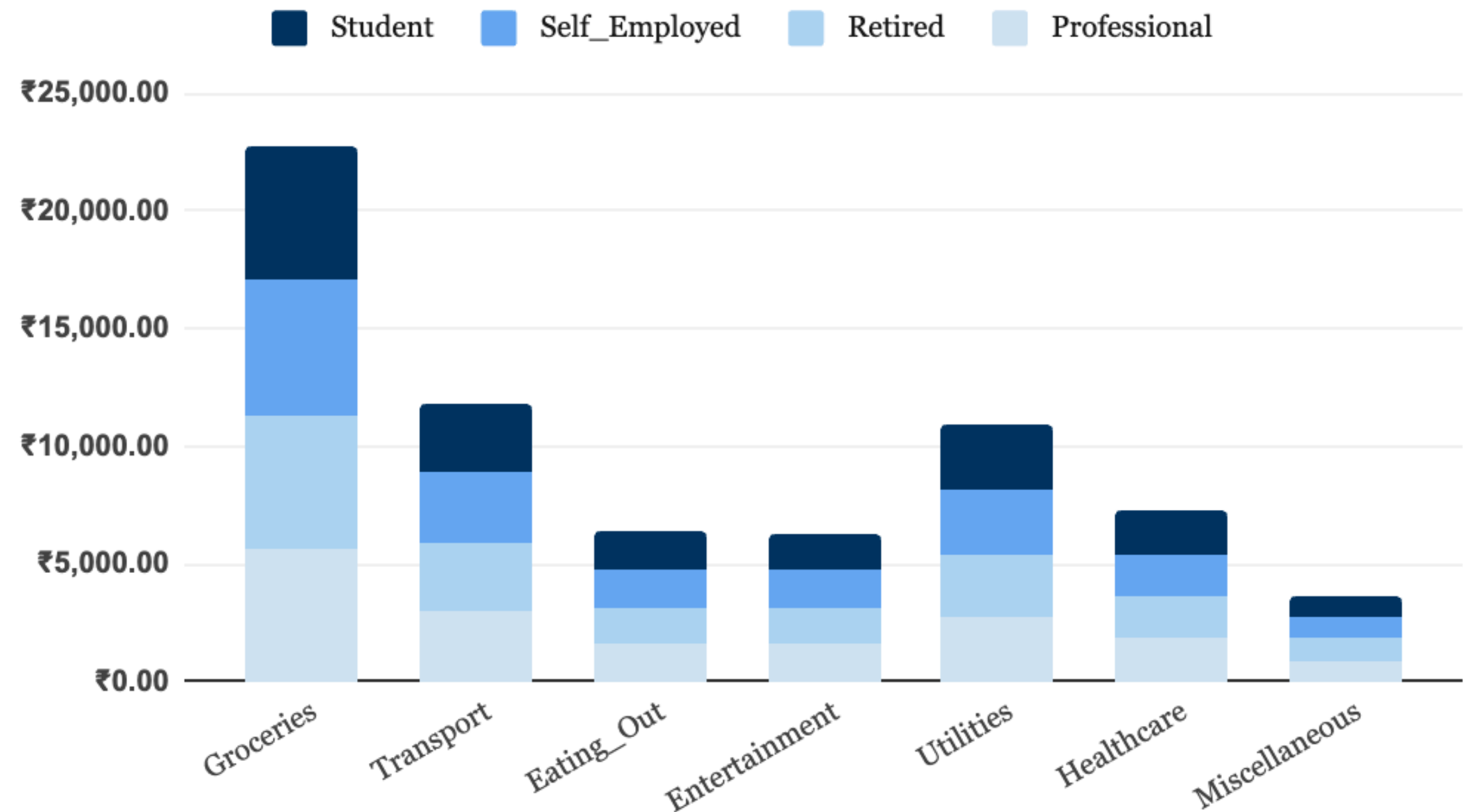
The dataset contains 20,000 financial records and is categorized across 4 base occupations and 3 city tiers.

- The overall average desired savings percentage across all groups is 9.80%
- The largest average spending category is Groceries at ₹5,205.67
- Fixed Expenses (Average Rent, Loan Repayment, and Insurance) account for approximately 30.35% of the average income.

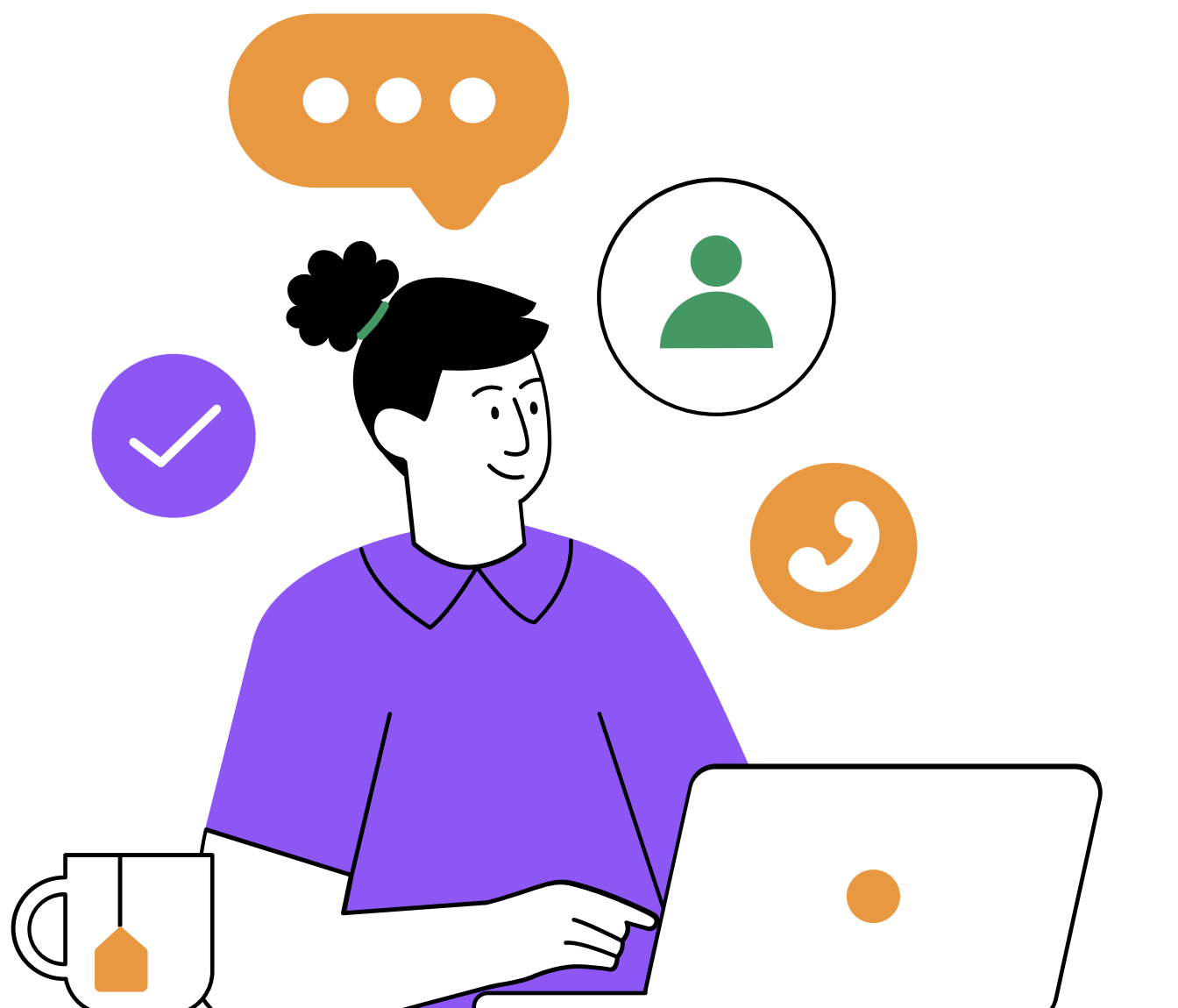
Q: Where does the average person's income go?



Average Monthly Expenditure by Occupation



Q: Can people actually meet their savings goals?



₹820

**Average monthly savings gap
(desired vs. disposable income)**

**The Average desired monthly
savings across all groups is
₹4,982.88.**

9.82% is the approximate
average desired savings rate for
the Low and Medium Income
Groups.

The Average desired savings
rate for the High Income Group
is approximately
9.81%.

KPI's And Matrix

Savings Rate (%)

How: $\text{Desired_Savings} / \text{Income} \times 100$

Why: Primary measure of financial health; tells us if goals are achievable.

Rent Burden (%)

How: $\text{Rent} / \text{Income} \times 100$

Why: Flags housing stress; >30% is internationally recognised as 'cost-burdened'.

Spending Category Mix

**How: $\text{Category} / \text{Income} \times 100$
per person**

Why: Reveals proportional behaviour across income groups and demographics.

Savings Gap (₹)

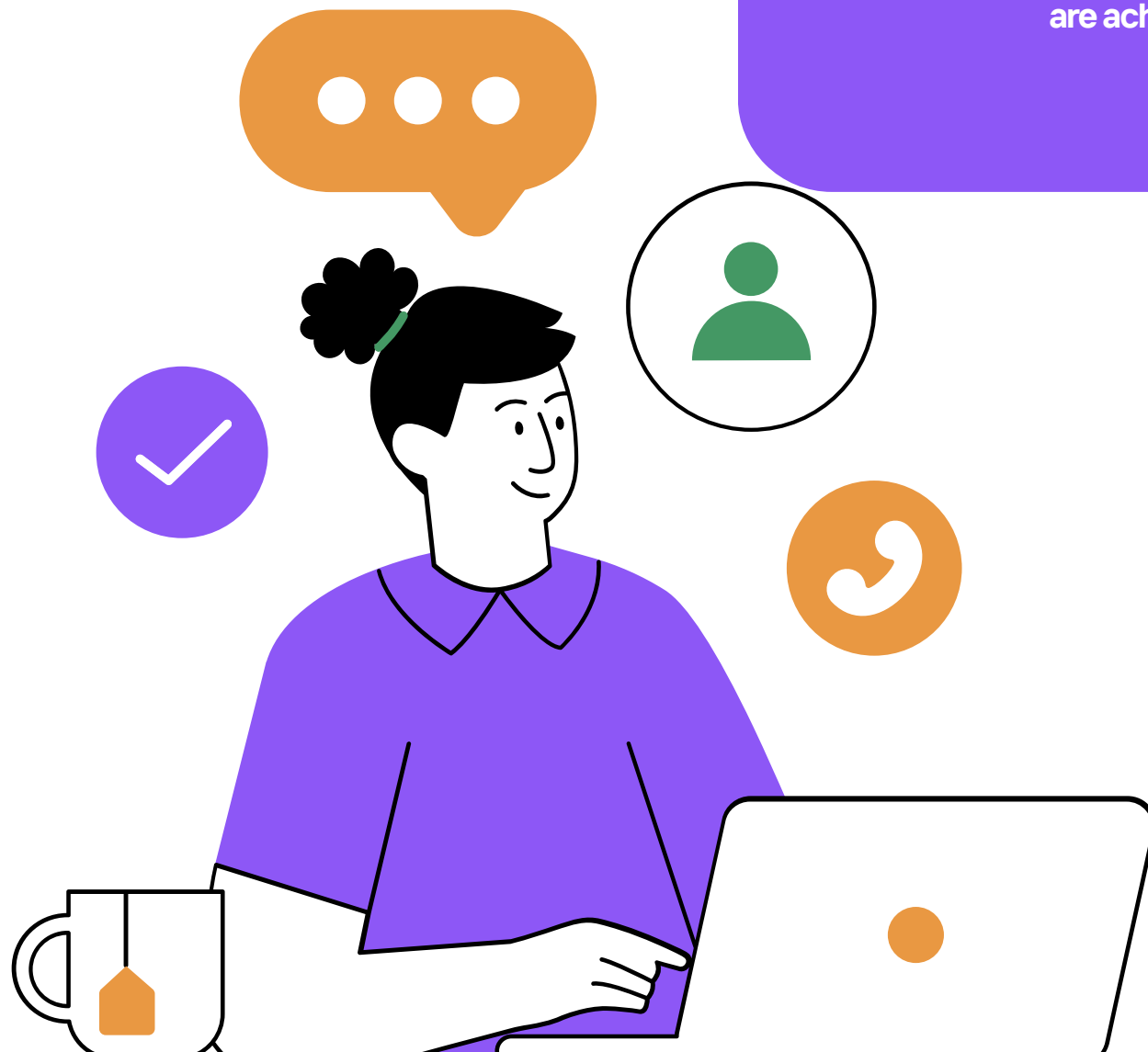
How: $\text{Desired_Savings} - \text{Disposable_Income}$

Why: Quantifies the exact shortfall each person faces monthly.

Savings Rate (%)

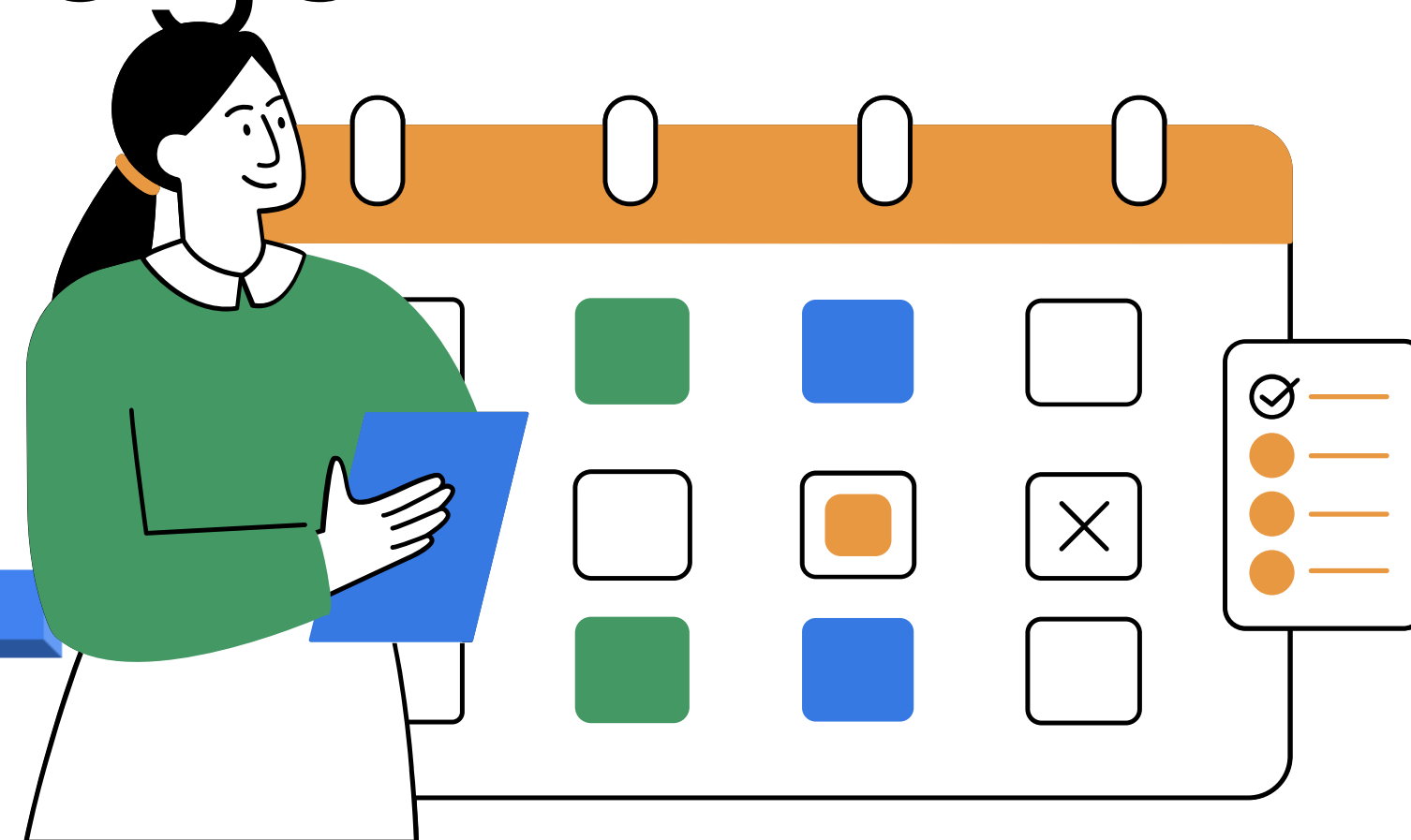
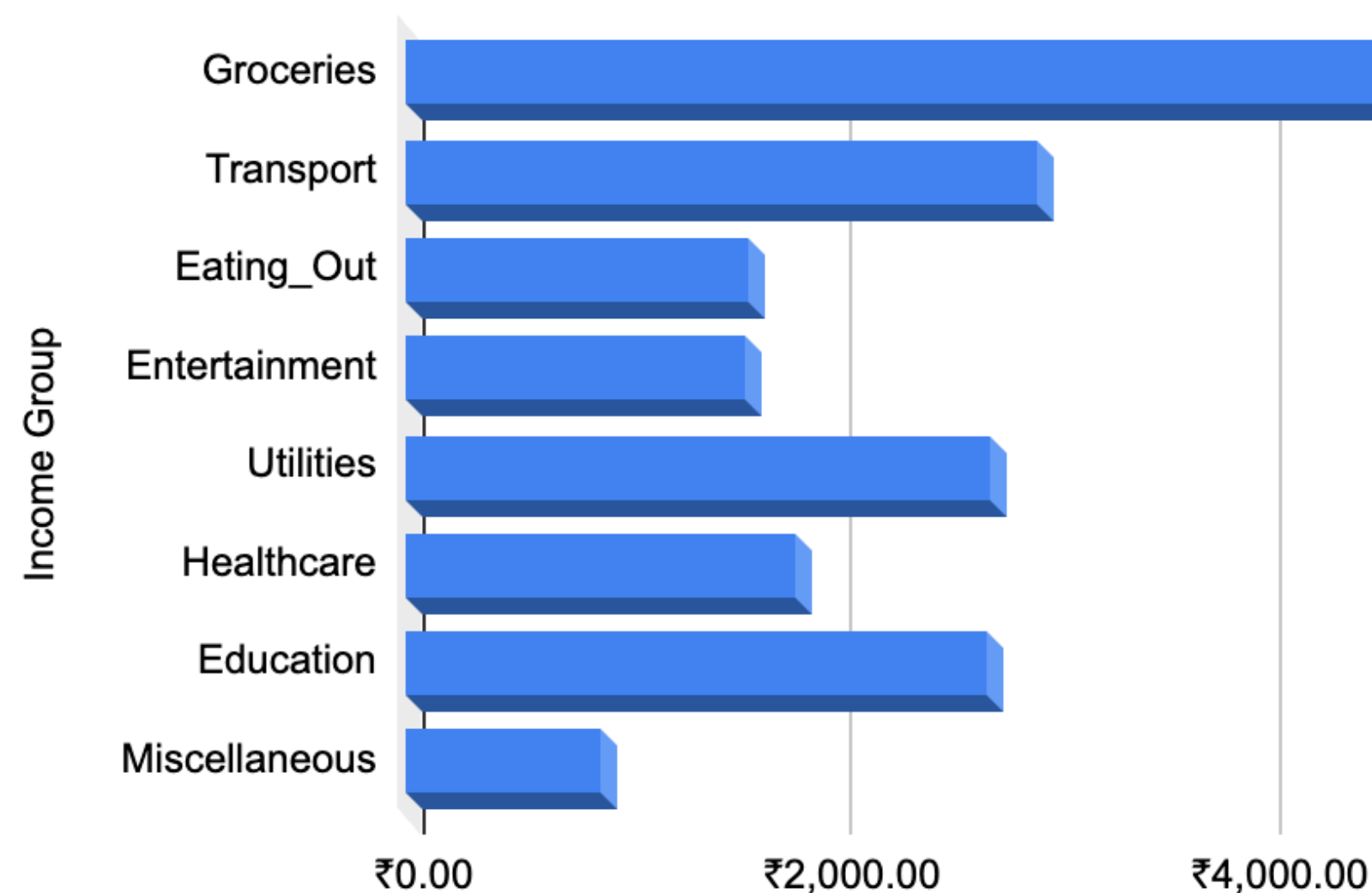
How: $\text{Desired_Savings} / \text{Income} \times 100$

Why: Primary measure of financial health; tells us if goals are achievable.



Average Monthly Budget Allocation

Avg Potential Savings by Category & Income Group

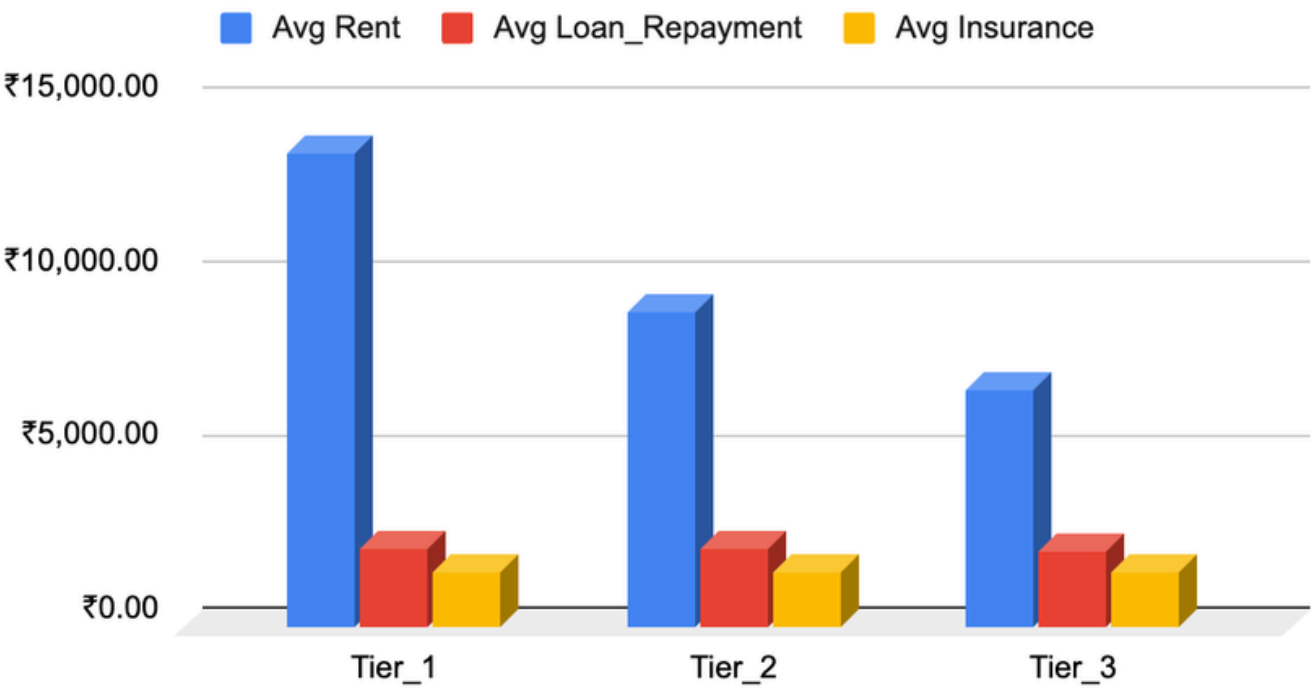


Budget Breakdown:

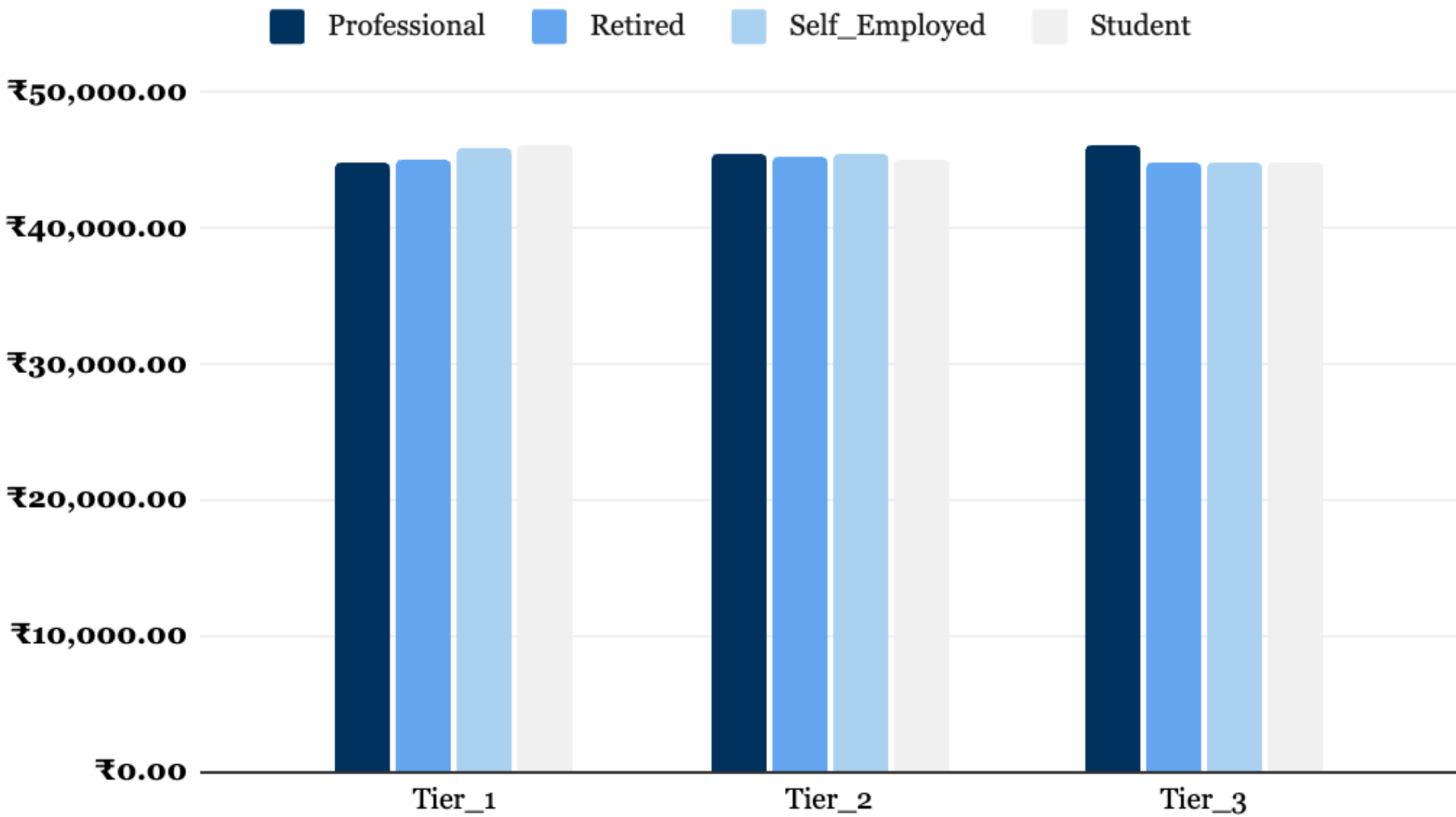
- 50-30-20 rule recommendation: 50% needs, 30% wants, 20% savings
- **Current allocation suggests opportunity for optimization**
- Fixed Expenses: **30.3%** (Rent, Loan, Insurance) Can go Upto 50%
- Variable Expenses: **59.85%** (Daily living costs) should be at 30 as per 50 - 30 -20 Rule.
- Savings: Current Saving is **9.80%** Target is 20%

Q: Does city type or occupation determine income?

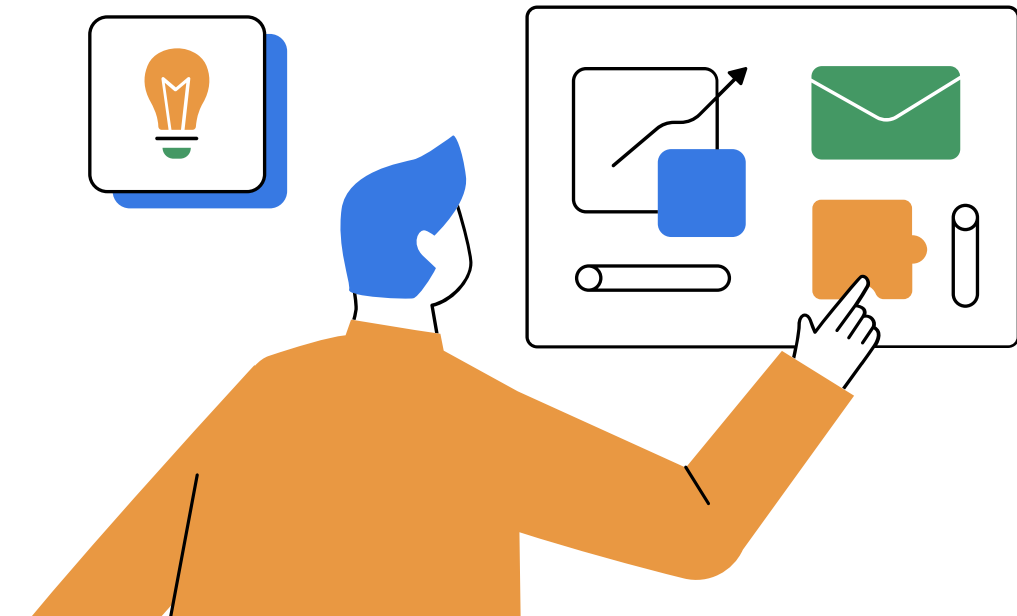
Rent & Loan Burden by City Tier



Average Income by Avg Income by Occupation & City Tier



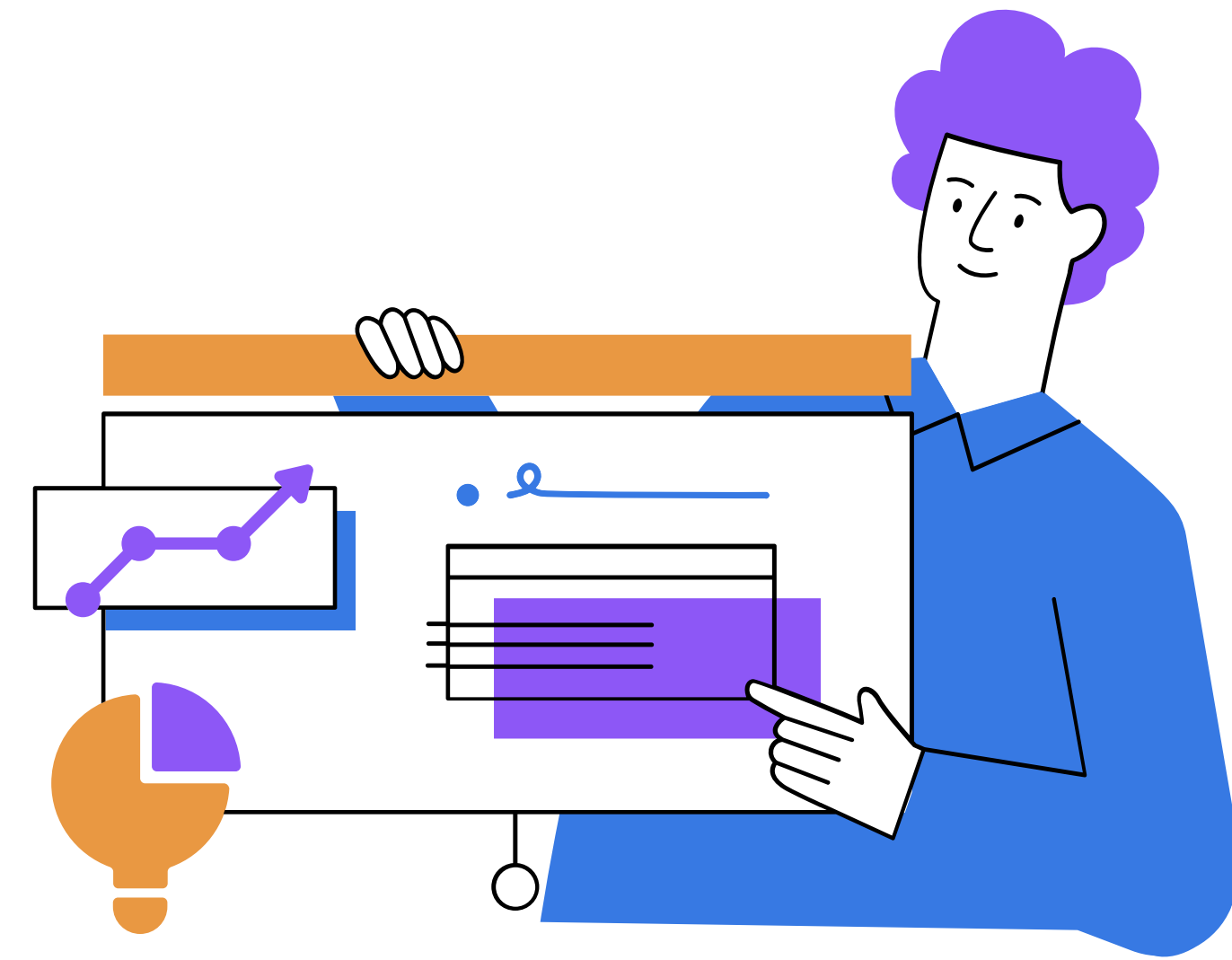
Surprising Finding: Income is nearly identical across all city tiers and occupations (₹40.7K–₹42K range). This means financial behaviour — not location or job type — is the real driver of savings outcomes.



Strategic Insights & Recommendations

Immediate Action Items:

1. **Implement grocery budgeting tools** → Target the largest average spending category (₹5,205.67 Groceries) for optimization.
2. **Launch city-specific financial literacy programs** → Target the Medium Income Group with the lowest average desired savings rate (9.77%) and the Low Income Group (9.87%) for the highest impact.
3. **Create transport optimization guides** → Focus on the second largest variable expense with an average of ₹2,704.47 (Transport).
4. **Develop utility efficiency programs** → Address the utility cost, which averages ₹2,504.98.
5. **Introduce automated savings plans** → Focus on raising the current average desired savings rate from 9.80% towards the 20% recommended by the 50/30/20 rule.
6. **Conduct quarterly spending reviews** → Maintain awareness and accountability.

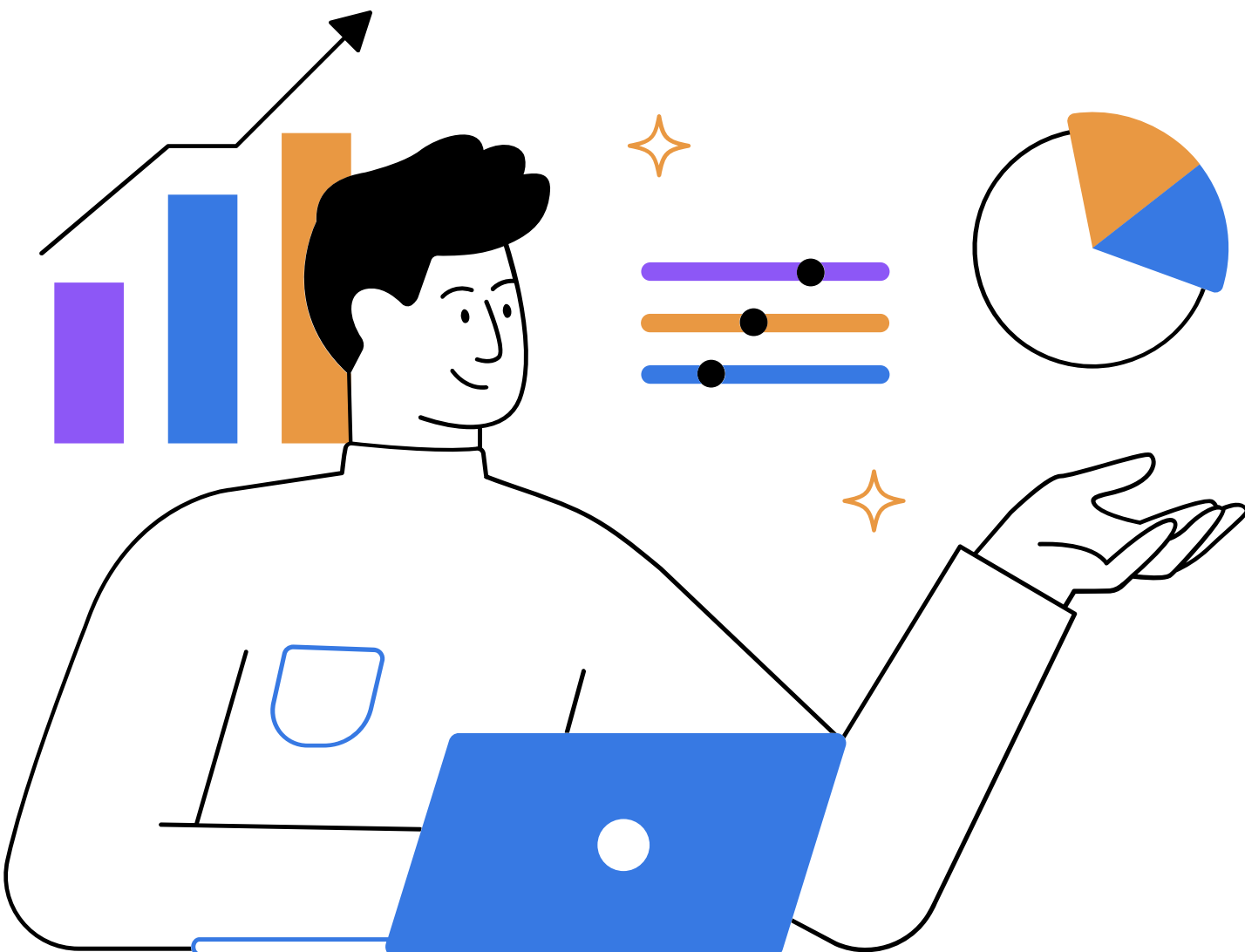


Critical Findings:

- Geography matters most: The overall average desired savings rate is 9.80% (Breakdown by City Tier is not available).
- Income drives all: Correlation data is not available between income and expenses.
- Lifestyle inflation: Data is not available to compare the proportional spending of higher earners.
- Groceries dominate: 12.52% of income, optimization potential is not verifiable with the current data.

How can individuals optimize spending across expense categories to increase savings without cutting essentials?

1. Reduce Grocery Spending: Target the largest expense (₹5,205.67) through meal planning or budgeting apps.
2. Reallocate Variable Budget: Shift funds from the excessive 59.85% "Remaining Budget" to savings.
3. Automate Savings: Immediately set up an automated transfer to raise the current 9.80% savings rate to the 20% goal.
4. Target Medium Income Group: Focus financial education on this group (lowest savings rate at 9.77%).
5. Optimize Second-Tier Expenses: Review Transport (₹2,704.47) and Utilities (₹2,504.98) for smaller, cumulative savings.



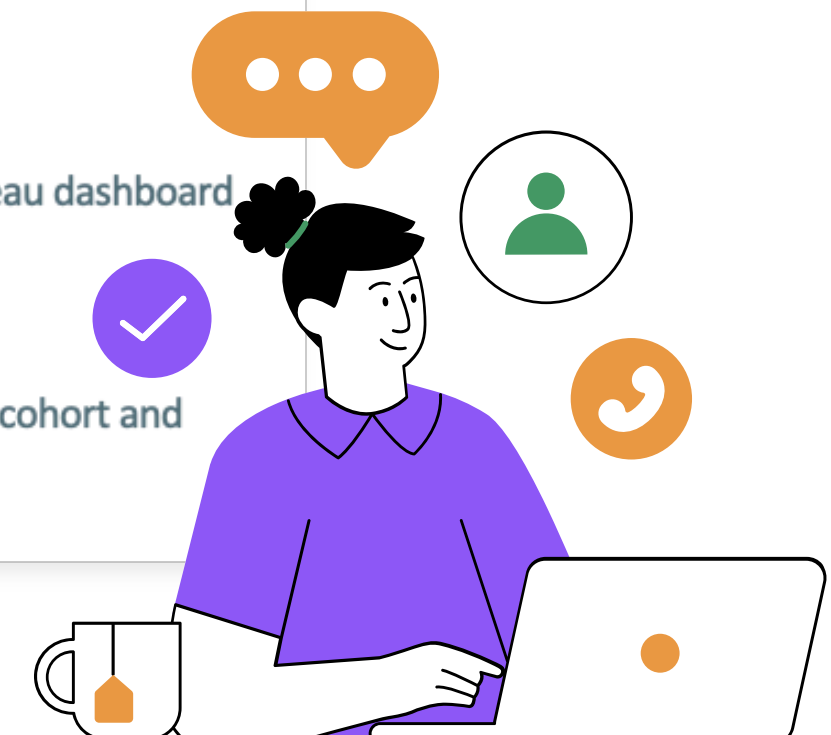
Limitations and Next Steps

⚠ Limitations & Data Gaps

- **Cross-sectional snapshot**
No longitudinal data — we cannot track if savings behaviour changes over time.
- **No actual savings data**
Dataset captures desired savings, not actual savings achieved — gap may be larger.
- **Income outliers**
Max income of ₹10.8L vs avg ₹41.5K suggests potential data entry errors or extreme cases.
- **No demographic depth**
Gender, education level, and marital status are absent — limits segmentation accuracy.
- **Potential savings model**
Potential savings are estimated benchmarks — actual realisable savings will vary by individual.

🚀 Next Steps & Improvements

- **Longitudinal study**
Track same individuals monthly to measure actual savings behaviour change over time.
- **ML-based savings predictor**
Train a regression model to predict achievable savings rate from spending patterns.
- **Add demographic variables**
Enrich dataset with gender, education, city name — enables sharper segmentation.
- **Live dashboard deployment**
Convert static analysis into an interactive Power BI / Tableau dashboard for real-time monitoring.
- **Pilot with fintech partner**
Test transport + grocery recommendations on a real user cohort and measure savings uplift.



Thank You

